Choosing the Right Control Structure

Student Handout

Name: _____ Date: _____

Instructions

Read each scenario and pick the control structure that fits it best. Write the letter of your answer, then write one sentence explaining why. Some scenarios could be solved more than one way; pick the clearest, most natural fit.

Option	Best used when
A. for-each loop	You visit every item in a collection, in order, and don't need the index
B. while loop	You repeat until a condition changes, and the body might run zero times
C. do-while loop	You repeat until a condition changes, and the body must run at least once
D. for loop	You know the number of repetitions, or you need a counter or index
E. if statement	You make a one-time decision, with no repetition
F. None of the above	No decision or repetition is needed; plain statements in sequence do the job

Scenarios

1. Shopping cart total. An online store keeps the items in a customer's cart in an `ArrayList<Item>`. Add up the price of every item to get the cart total.

Answer: _____ Why: _____

_____

2. Game menu. A text game shows a menu (Play, Settings, Quit), reads the player's choice, and acts on it. The menu keeps reappearing until the player picks Quit. The menu must be shown at least once.

Answer: _____ Why: _____

_____

3. Free shipping. After checkout, the program prints "You qualify for free shipping!" if the order total is $50 or more.

Answer: _____ Why: _____

_____

4. Reverse order. Print the elements of an `int[] scores` array from the last element to the first.

Answer: _____ Why: _____

_____

5. Reading a file. A program reads lines from a text file and prints each one until there are no lines left. The file might be empty.

Answer: _____ Why: _____

_____

6. Swap two values. Two variables, a and b, hold numbers. Exchange their values so a holds what was in b and b holds what was in a.

Answer: _____ Why: _____

_____

7. Email blast. A club stores its members' email addresses in a list. Print every address so the secretary can copy them into an email.

Answer: _____ Why: _____

_____

8. Roll until six. Simulate rolling a die and print each roll. Keep rolling until a 6 comes up. You can't know the result until you've rolled at least once.

Answer: _____ Why: _____

_____

9. Doubling savings. A savings account starts at $1,000 and earns 5% interest per year. Count how many years it takes for the balance to reach $2,000.

Answer: _____ Why: _____

_____

10. Rocket countdown. Print a countdown from 10 down to 1, then print "Liftoff!"

Answer: _____ Why: _____

_____